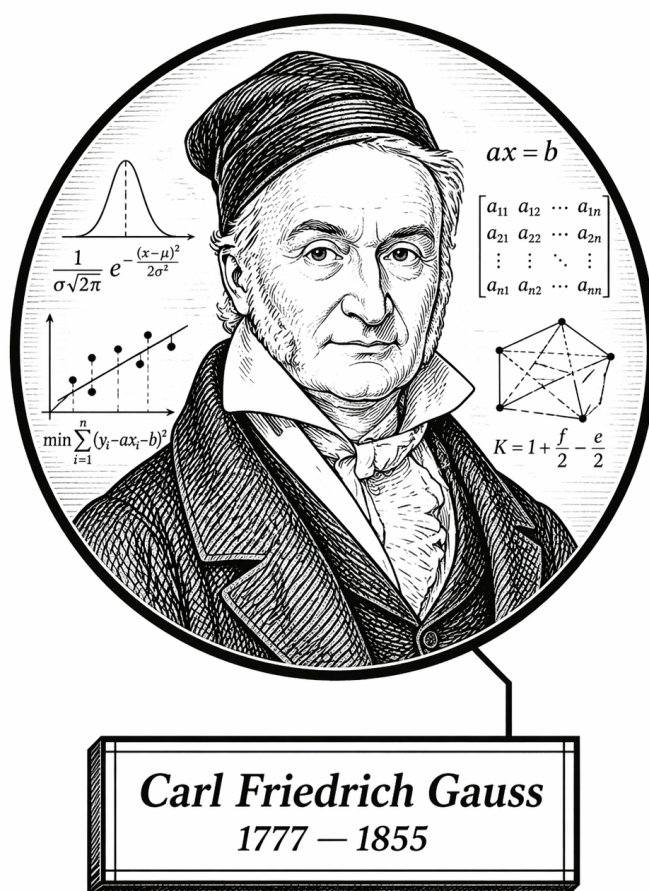


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Volume V

Topology and Geometry



"So many squares..."

– Carl Gauss (probably)

Volume V

Modern Analysis

Self-Study Notes

William Sollers

June 25, 2026

Part I

Modern Analysis

Chapter 1

Measure Theory

measure-theory

Functional Analysis $\rightarrow$ **Measure Theory** $\rightarrow$ Probability Theory

1.1 Measure Theory Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Measure Theory, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 2

Probability Theory

probability-theory

Measure Theory $\rightarrow$ **Probability Theory** $\rightarrow$ Quantum Calculus

2.1 Probability Theory Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Probability Theory, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 3

Functional Analysis

functional-analysis

Complex Analysis $\rightarrow$ **Functional Analysis** $\rightarrow$ Measure Theory

3.1 Functional Analysis Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Functional Analysis, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 4

Complex Analysis

complex-analysis

Real Analysis $\rightarrow$ **Complex Analysis** $\rightarrow$ Functional Analysis

4.1 Complex Analysis Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Complex Analysis, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Bibliography